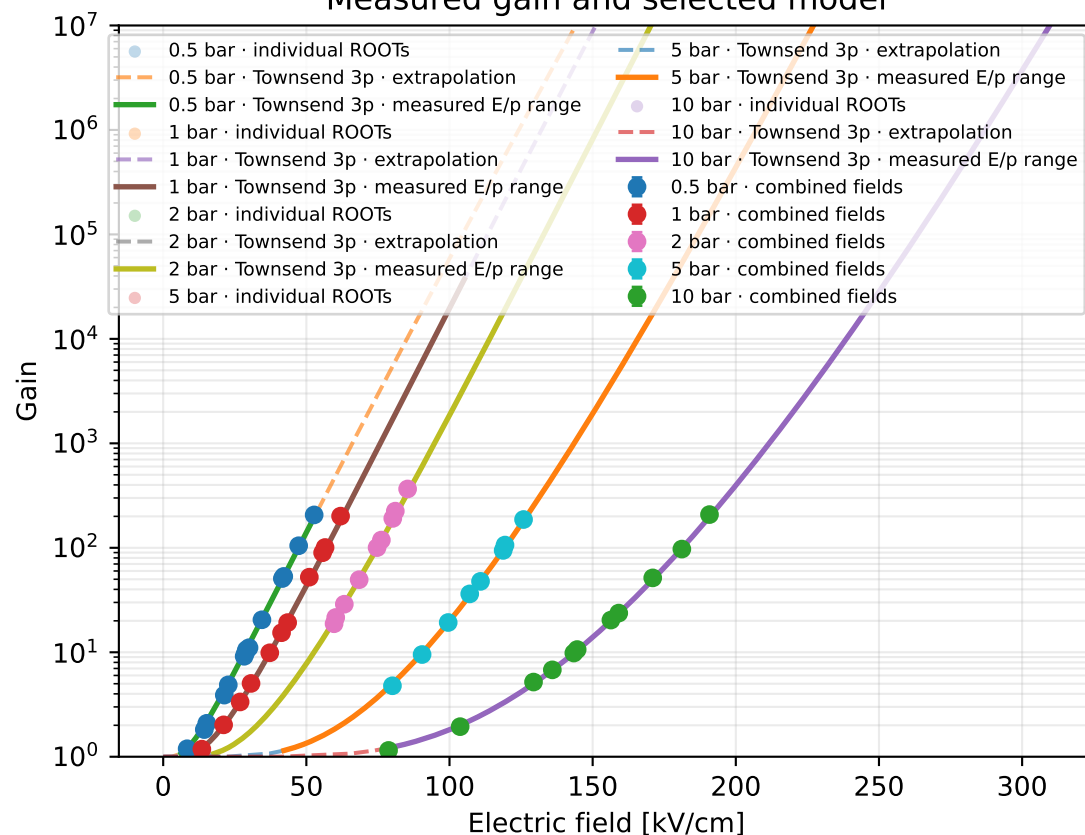
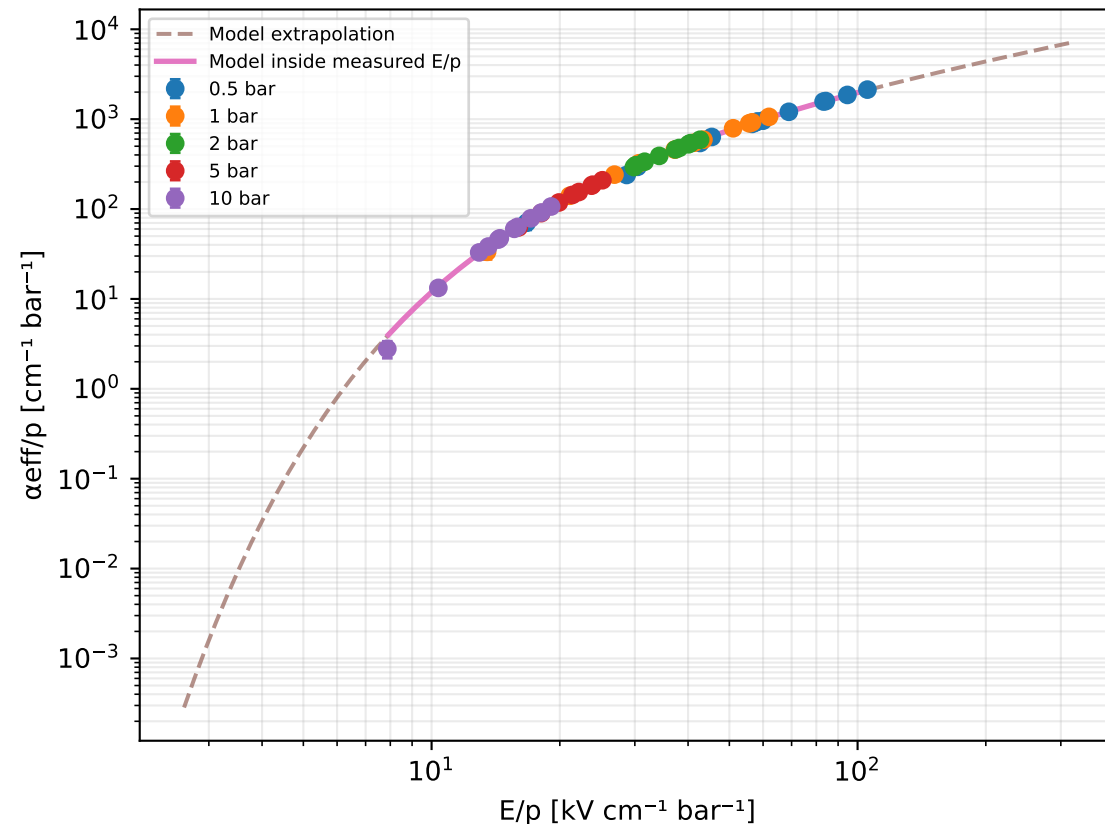


# ArCF4 · Ar 99.9 % / CF4 0.1 % · gap 0.05 mm

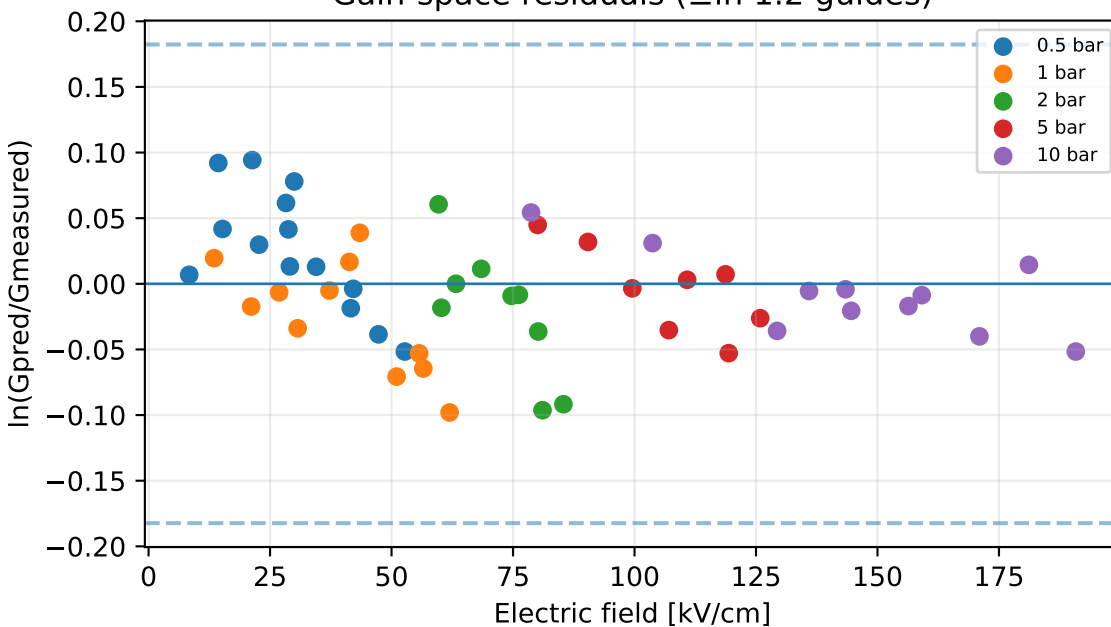
## Measured gain and selected model



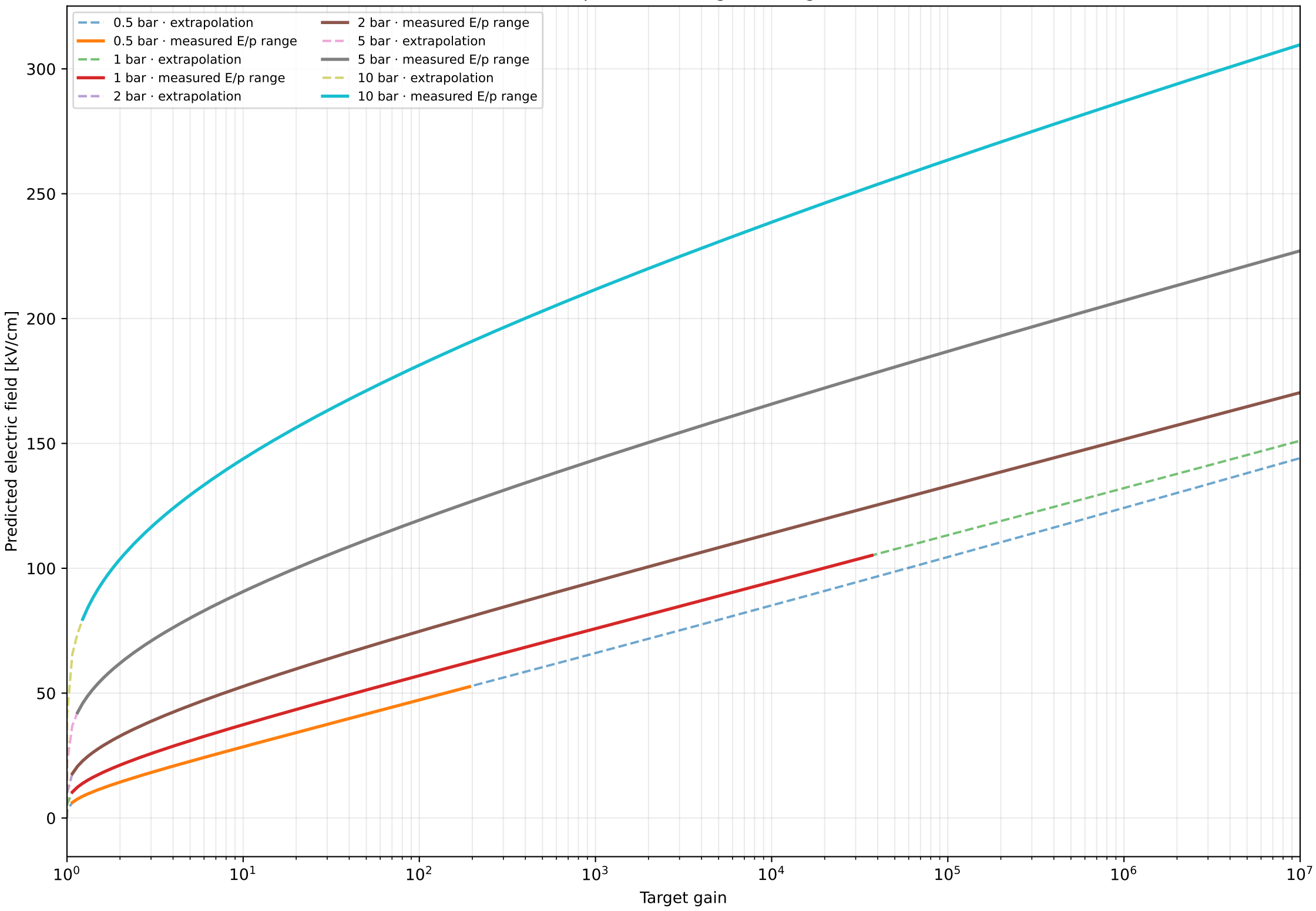
## Reduced Townsend fit



## Gain-space residuals ( $\pm \ln 1.2$ guides)



ROOT runs: 76  
 Unique physical points: 53  
 Pressures: 0.5, 1, 2, 5, 10 bar  
 Selected model: Townsend 3p  
 Status: accepted  
 Training RMSE:  $\times 1.057$   
 Cross-validation median:  $\times 1.022$   
 Cross-validation worst:  $\times 1.101$   
 Reduced-field range: 7.871–105.5 kV cm $^{-1}$  bar $^{-1}$   
 $A=41.905$ ,  $B=33.525$ ,  $m=0.90969$ ,  $n=1$   
 Covariance condition: 1.37e+04

Field extrapolation from gain 1 to gain  $10^7$ 

| p [bar] | E [kV/cm] | E/p     | Gain     | σGain  | αeff [cm <sup>-1</sup> ] | npe  | runs |
|---------|-----------|---------|----------|--------|--------------------------|------|------|
| 0.5     | 8.36163   | 16.7233 | 1.192593 | 0.0357 | 35.2259                  | 135  | 1    |
| 0.5     | 14.3529   | 28.7058 | 1.819728 | 0.0545 | 119.737                  | 294  | 1    |
| 0.5     | 15.222    | 30.4439 | 2.089918 | 0.0626 | 147.425                  | 367  | 1    |
| 0.5     | 21.342    | 42.684  | 3.889688 | 0.117  | 271.666                  | 417  | 1    |
| 0.5     | 22.7351   | 45.4703 | 4.877358 | 0.146  | 316.921                  | 424  | 1    |
| 0.5     | 28.2833   | 56.5666 | 9.152709 | 0.274  | 442.81                   | 406  | 1    |
| 0.5     | 28.7698   | 57.5397 | 9.904872 | 0.297  | 458.605                  | 431  | 1    |
| 0.5     | 29.0768   | 58.1536 | 10.57447 | 0.317  | 471.688                  | 423  | 1    |
| 0.5     | 29.9803   | 59.9606 | 11.06073 | 0.331  | 480.68                   | 494  | 1    |
| 0.5     | 34.4856   | 68.9713 | 20.47027 | 0.614  | 603.795                  | 370  | 1    |
| 0.5     | 41.6121   | 83.2243 | 50.79639 | 1.52   | 785.565                  | 388  | 1    |
| 0.5     | 42.1127   | 84.2254 | 53.2367  | 1.6    | 794.95                   | 376  | 1    |
| 0.5     | 47.3161   | 94.6321 | 104.6825 | 3.14   | 930.186                  | 274  | 1    |
| 0.5     | 52.7291   | 105.458 | 206.4757 | 6.18   | 1066.04                  | 288  | 1    |
| 1       | 13.4987   | 13.4987 | 1.181818 | 0.0353 | 33.4108                  | 143  | 1    |
| 1       | 21.1041   | 21.1041 | 2.019737 | 0.0605 | 140.593                  | 456  | 1    |
| 1       | 26.8704   | 26.8704 | 3.345528 | 0.1    | 241.525                  | 492  | 1    |
| 1       | 30.6796   | 30.6796 | 5.032922 | 0.151  | 323.2                    | 486  | 1    |
| 1       | 37.2266   | 37.2266 | 9.887821 | 0.296  | 458.261                  | 624  | 1    |
| 1       | 41.3391   | 41.3391 | 15.39231 | 0.461  | 546.774                  | 520  | 1    |
| 1       | 43.4688   | 43.4688 | 19.2407  | 0.577  | 591.406                  | 511  | 1    |
| 1       | 51.0405   | 51.0405 | 52.35269 | 1.57   | 791.601                  | 465  | 1    |
| 1       | 55.6766   | 55.6766 | 89.72667 | 3.48   | 899.354                  | 300  | 1    |
| 1       | 56.5355   | 56.5355 | 100.6879 | 3.01   | 922.405                  | 471  | 1    |
| 1       | 61.954    | 61.954  | 201.0167 | 7.12   | 1060.68                  | 300  | 1    |
| 2       | 59.6795   | 29.8398 | 18.75246 | 0.562  | 586.265                  | 711  | 1    |
| 2       | 60.2611   | 30.1305 | 21.53224 | 0.645  | 613.91                   | 729  | 1    |
| 2       | 63.2603   | 31.6302 | 28.84304 | 0.25   | 672.374                  | 7792 | 12   |
| 2       | 68.4502   | 34.2251 | 49.5378  | 1.48   | 780.547                  | 582  | 1    |
| 2       | 74.7029   | 37.3515 | 100.4216 | 3.01   | 921.875                  | 593  | 1    |

| p [bar] | E [kV/cm] | E/p     | Gain     | $\sigma$ Gain | $\alpha_{\text{eff}}$ [cm <sup>-1</sup> ] | npe  | runs |
|---------|-----------|---------|----------|---------------|-------------------------------------------|------|------|
| 2       | 76.2031   | 38.1016 | 118.6333 | 5.42          | 955.208                                   | 300  | 1    |
| 2       | 80.1889   | 40.0944 | 191.25   | 7.85          | 1050.72                                   | 300  | 1    |
| 2       | 81.0671   | 40.5336 | 224.3933 | 9.48          | 1082.68                                   | 300  | 1    |
| 2       | 85.3679   | 42.684  | 365.777  | 11            | 1180.4                                    | 601  | 1    |
| 5       | 80.0676   | 16.0135 | 4.781213 | 0.143         | 312.939                                   | 841  | 1    |
| 5       | 90.425    | 18.085  | 9.51861  | 0.285         | 450.65                                    | 806  | 1    |
| 5       | 99.5451   | 19.909  | 19.27771 | 0.578         | 591.79                                    | 839  | 1    |
| 5       | 107.062   | 21.4124 | 36.21266 | 1.09          | 717.882                                   | 790  | 1    |
| 5       | 110.833   | 22.1666 | 47.72716 | 1.43          | 773.1                                     | 799  | 1    |
| 5       | 118.739   | 23.7478 | 94.435   | 3.1           | 909.582                                   | 600  | 1    |
| 5       | 119.386   | 23.8771 | 106.2383 | 3.42          | 933.137                                   | 600  | 1    |
| 5       | 125.863   | 25.1726 | 186.576  | 1.77          | 1045.77                                   | 7800 | 13   |
| 10      | 78.7146   | 7.87146 | 1.149378 | 0.0344        | 27.8441                                   | 241  | 1    |
| 10      | 103.718   | 10.3718 | 1.941281 | 0.0582        | 132.67                                    | 562  | 1    |
| 10      | 129.343   | 12.9343 | 5.187256 | 0.155         | 329.241                                   | 769  | 1    |
| 10      | 135.914   | 13.5914 | 6.786885 | 0.204         | 382.998                                   | 732  | 1    |
| 10      | 143.422   | 14.3422 | 9.833333 | 0.295         | 457.156                                   | 828  | 1    |
| 10      | 144.589   | 14.4589 | 10.62118 | 0.318         | 472.57                                    | 916  | 1    |
| 10      | 156.373   | 15.6373 | 20.33732 | 0.61          | 602.492                                   | 919  | 1    |
| 10      | 159.111   | 15.9111 | 23.71723 | 0.711         | 633.24                                    | 923  | 1    |
| 10      | 170.98    | 17.098  | 51.52761 | 1.55          | 788.424                                   | 851  | 1    |
| 10      | 181.153   | 18.1153 | 97.22167 | 3.38          | 915.399                                   | 600  | 1    |
| 10      | 190.807   | 19.0807 | 208.01   | 10.6          | 1067.52                                   | 300  | 1    |

Progressive model selection

The higher-order model is selected only when new points improve out-of-sample prediction.

| Candidate   | k | AICc | CV median | CV worst | usable | warnings                       |
|-------------|---|------|-----------|----------|--------|--------------------------------|
| Townsend 2p | 2 | -179 | ×1.24     | ×3.15    | no     | large cross validation outlier |
| Townsend 3p | 3 | -301 | ×1.02     | ×1.1     | yes    | —                              |
| Townsend 4p | 4 | -326 | ×1.02     | ×1.14    | yes    | —                              |